SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSA47 – DEEP LEARNING
ASSIGNMENT – 1

CO1 – Mathematical Foundations and Machine Learning Principles

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Slot B

TASK 1 – Patient Data Representation and Normalization

a) Effect of using raw features

The four features have very different numerical scales. Age may range roughly from 0 to 100, while blood pressure, glucose and cholesterol are measured on different ranges. If raw values are used, a feature with a larger numerical range can dominate distance calculations and may produce disproportionately large gradients during training. As a result, the model may give importance to scale rather than to the actual medical relevance of a feature. Optimization can also become slower or less stable.

b) Feature matrix and normalization

Consider the following illustrative data for three patients. The columns are age (years), blood pressure (mmHg), glucose (mg/dL) and cholesterol (mg/dL).

Patient	Age	Blood pressure	Glucose	Cholesterol
P1	20	110	90	180
P2	40	130	120	200
P3	60	150	150	220

The feature matrix is:

X = [[20, 110, 90, 180], [40, 130, 120, 200], [60, 150, 150, 220]].

Use min–max normalization for each column: x' = (x − minimum) / (maximum − minimum). This maps each feature to the range 0 to 1.

Patient	Age'	BP'	Glucose'	Cholesterol'
P1	0.00	0.00	0.00	0.00
P2	0.50	0.50	0.50	0.50
P3	1.00	1.00	1.00	1.00

Verification: for P2 age, (40 − 20)/(60 − 20) = 20/40 = 0.5. For P2 blood pressure, (130 − 110)/(150 − 110) = 20/40 = 0.5. The same calculation gives 0.5 for glucose and cholesterol.

c) Importance of normalization

Distance-based methods such as K-nearest neighbours and K-means use distances between feature vectors. Normalization prevents a feature with a large range from dominating those distances. Gradient-based methods such as neural networks use gradients to update parameters; scaling inputs to comparable ranges can improve numerical conditioning, make optimization more stable and help the model converge faster.

TASK 2 – Vector Representation and Matrix Operations

a) Vector formulation

Let one patient's feature vector be $x = [\text{age, glucose, blood pressure, cholesterol}]^T$ and the importance-weight vector be $w = [w_1, w_2, w_3, w_4]^T$. The weighted preliminary risk score is the dot product $S = w^T x = w_1(\text{age}) + w_2(\text{glucose}) + w_3(\text{BP}) + w_4(\text{cholesterol})$.

b) Numerical example

Suppose the normalized feature vector is $x = [0.4, 0.8, 0.6, 0.5]^T$ and the weights are $w = [0.1, 0.4, 0.3, 0.2]^T$. Then:

$S = 0.1(0.4) + 0.4(0.8) + 0.3(0.6) + 0.2(0.5)$

$= 0.04 + 0.32 + 0.18 + 0.10 = 0.64.$

The preliminary weighted score is 0.64. It is a numerical illustration of the combined weighted features, not a clinically validated probability or diagnosis.

TASK 3 – Probability and Conditional Probability

Let D denote the event that a patient has the disease, and G denote the event that the patient has elevated glucose. Total patients = 1,000; diseased = 200; non-diseased = 800. Among diseased patients, 160 have elevated glucose; among non-diseased patients, 120 have elevated glucose.

a) Probability of elevated glucose

Total patients with elevated glucose = 160 + 120 = 280.

$P(G) = 280/1000 = 0.28 = 28\%.$

b) Probability of disease given elevated glucose

$P(D \mid G) = P(D \cap G) / P(G) = 160/280 = 4/7 \approx 0.5714 = 57.14\%.$

c) Probability of elevated glucose given disease

$P(G \mid D) = 160/200 = 0.80 = 80\%.$

d) Healthcare interpretation

Conditional probability helps estimate how common disease is among patients who show a particular indicator. Here, about 57.14% of patients with elevated glucose have the disease in this sample. This can help clinicians prioritize such patients for further assessment. It should not be used alone to diagnose disease; other symptoms, tests and clinical judgement are required.

TASK 4 – Bayesian Statistics for Disease Classification

a) Bayes' theorem calculation

Let D = disease and + = positive test. Given P(D) = 0.10, P(not D) = 0.90, P(+ | D) = 0.90 and P(+ | not D) = 0.05.

$P(D \mid +) = [P(+ \mid D) \times P(D)] / P(+)$.

$P(+) = P(+ \mid D)P(D) + P(+ \mid \text{not } D)P(\text{not } D)$

$= (0.90 \times 0.10) + (0.05 \times 0.90) = 0.09 + 0.045 = 0.135$.

Therefore, $P(D \mid +) = 0.09/0.135 = 0.6667 \approx 66.67\%$.

b) Prior and likelihood

The prior probability P(D) = 10% is the disease probability before considering the test result. The likelihood P(+ | D) = 90% describes how likely a positive result is when the patient has the disease. The posterior probability P(D | +) = 66.67% is the updated probability after observing a positive result.

c) Usefulness of Bayesian reasoning

Medical decisions often involve incomplete information and tests that are not perfectly accurate. Bayesian reasoning combines existing knowledge (the prior) with new evidence (the likelihood) to update the probability of disease. It makes uncertainty explicit and helps clinicians interpret test results in the context of disease prevalence.

TASK 5 – Entropy and Information Gain

a) Entropy calculation

There are 40 high-risk and 60 low-risk records out of 100. Thus p(high) = 0.4 and p(low) = 0.6.

$H(Y) = -\Sigma\, p_i \log_2(p_i) = -[0.4 \log_2(0.4) + 0.6 \log_2(0.6)]$.

$H(Y) = -[0.4(-1.3219) + 0.6(-0.7370)] \approx 0.971$ bits.

b) Meaning of entropy

Entropy measures uncertainty or impurity in the class labels. An entropy of approximately 0.971 bits is relatively close to the maximum of 1 bit for two equally likely classes, indicating substantial uncertainty in this 40:60 split.

c) Information gain

Information gain is the reduction in entropy after splitting records using a feature. For glucose, calculate the weighted entropy of the groups formed by a glucose threshold and subtract it from the original entropy: IG = H(before) − H(after). A glucose split with higher information gain separates high-risk and low-risk records more effectively and may be selected by a decision tree.

TASK 6 – Numerical Computation and Distance Measurement

a) Euclidean distance

Patient A = (0.70, 0.80, 0.60) and Patient B = (0.50, 0.75, 0.90).

$d(A,B) = \sqrt{[(0.70-0.50)^2 + (0.80-0.75)^2 + (0.60-0.90)^2]}$

$= \sqrt{[0.20^2 + 0.05^2 + (-0.30)^2]}$

$= \sqrt{[0.0400 + 0.0025 + 0.0900]} = \sqrt{0.1325} \approx 0.364.$

b) Similarity interpretation

A smaller Euclidean distance means the two normalized feature vectors are more similar across the selected measurements. A larger distance indicates greater difference. The value 0.364 represents their combined separation in the normalized three-feature space.

c) Need for normalization

Glucose, blood pressure and cholesterol have different units and numerical ranges. Without normalization, the feature with the largest scale could dominate the distance. Normalization makes the contributions more comparable before measuring similarity.

TASK 7 – Bias–Variance Trade-off

a) Identify the models

- Model A: High training error and high validation error indicate high bias (underfitting).
- Model B: Very low training error but high validation error indicates high variance (overfitting).
- Model C: Low and similar training and validation errors indicate good generalization and a suitable bias–variance balance.

b) Effect of model complexity

A model with very low complexity may be unable to represent important patterns, producing high bias. Increasing complexity usually reduces bias because the model can fit more complex relationships. However, excessive complexity can make the model fit noise in the training data, increasing variance and worsening performance on unseen data.

c) Improvements

For Model A, use a more expressive model, add relevant features, or reduce excessive regularization. For Model B, use regularization, simplify the model, collect more representative training data, apply cross-validation and use early stopping where appropriate.

TASK 8 – Overfitting, Underfitting and Model Capacity

a) Classification

- The simple model that fails to capture disease patterns is underfitting.
- The moderately complex model that performs consistently is appropriately generalized.
- The highly complex model that performs almost perfectly on training data but poorly on unseen patients is overfitting.

b) Model capacity

Model capacity is the range or complexity of relationships a model can learn. Low-capacity models may miss complex relationships. Higher-capacity models can represent more detailed patterns, but they require sufficient representative data and suitable regularization.

c) Why increasing capacity can eventually overfit

At first, increasing capacity allows the model to learn meaningful patterns that a simple model could not capture, so performance improves. Beyond a suitable point, the model may also learn random fluctuations and noise in the training set. Training error continues to fall, but error on unseen patients may rise. This is overfitting.

TASK 9 – Supervised Learning and Model Evaluation

a) Why supervised learning

This is a supervised learning problem because historical patient records contain input features together with known labels: diseased or disease-free. The model learns a relationship between the features and labels to predict outcomes for new patients.

b) Training and testing sets

The training set is used to learn model parameters and patterns. The testing set is kept separate and used after training to estimate how well the model performs on unseen records. The test set should not be used to tune the model, because that would make the evaluation overly optimistic.

c) Accuracy

Accuracy = (number of correct predictions / total predictions) × 100 = (92/100) × 100 = 92%.

The model correctly classified 92 out of 100 test patients. Accuracy alone may hide poor performance on a minority class, so measures such as sensitivity, specificity, precision, recall and the confusion matrix should also be examined.

TASK 10 – Unsupervised Learning and Patient Grouping

a) Why unsupervised learning

The records contain patient features but no predefined risk labels. Unsupervised learning is suitable because it can identify structure or groups in the data without known target labels.

b) K-means procedure

K-means can group patients as follows: (1) choose the number of clusters K; (2) initialize K centroids; (3) assign each patient to the nearest centroid using a distance measure; (4) recompute each centroid as the mean of the points assigned to it; and (5) repeat assignment and centroid updates until the assignments or centroids stabilize.

c) Distance and similarity

Distance measures the separation between patient feature vectors. Patients with smaller distances are more similar in the selected features and are more likely to be assigned to the same cluster. Features should be normalized before distance-based clustering.

d) Clinical use

Clusters may reveal groups with similar age, blood pressure, glucose and cholesterol profiles. Clinicians can investigate these groups further, identify patterns for additional testing or research, and plan follow-up. A cluster is not a diagnosis and must not replace professional medical assessment.

TASK 11 – Comparison of Supervised and Unsupervised Learning

a) Suitable paradigm

Dataset A has patient features and disease labels, so supervised learning is suitable. Dataset B has features without labels, so unsupervised learning is suitable for discovering groups or patterns.

b) Objectives and outputs

Aspect	Supervised learning	Unsupervised learning
Input	Features with known labels	Features without labels
Objective	Learn to predict a target label	Discover structure or groups
Typical output	Predicted class or probability	Clusters, associations or patterns
Example	Disease-risk classification	K-means patient grouping

c) Recommendation for a new patient's disease status

Use a supervised classification model trained on Dataset A because the task is to predict a known outcome—diseased or disease-free. The model can produce a class or estimated probability for a new patient. Dataset B can additionally be clustered to discover patient subgroups and may provide useful exploratory insights, but clustering alone does not directly provide disease labels.

TASK 12 – Design Challenge: Selecting and Adapting a Learning Strategy

A. Impact Analysis

a) Increasing the dataset from 10,000 to 1,000,000 records increases storage, memory use and computation. The feature matrix has one million rows, so data loading, preprocessing, model training and inference must be planned efficiently. Batch processing, efficient data types, vectorized operations and scalable storage can reduce resource demands. Training time may increase, although the larger dataset can improve learning if it is representative.

b) Normalization is needed because the features use different scales and distance or gradient calculations can otherwise be dominated by large-valued features. Missing values must be identified and handled using a justified method, such as median imputation for numerical features, while retaining missingness indicators when useful. Data variability matters because the model must represent different patient groups and avoid learning from a narrow or biased sample.

B. Evaluation of the Original Approach

a) A single supervised model is not sufficient for both objectives. Supervised classification uses labels to predict disease risk, whereas patient-group discovery requires identifying structure in records without relying on predefined labels. A classifier does not automatically discover meaningful unlabeled groups.

b) Labelled learning cannot directly use the target labels that are absent from unlabelled records. Ignoring those records may discard useful information about patient subgroups and feature distributions. A supervised model may also be limited by the quantity, quality and representativeness of available labels.

C. Adaptation or Redesign

a) Use a combined strategy. First, clean the data, check missingness and normalize numerical features. Train a supervised classifier on labelled records to predict disease status and risk probability. Separately, apply K-means or another suitable clustering method to normalized patient features, including eligible unlabelled records, to

discover patient groups. Compare cluster profiles with clinical knowledge and use them to guide further investigation. Keep clinical diagnosis with qualified healthcare professionals.

b) Incorporate probability by calibrating and interpreting the classifier's predicted disease-risk probabilities, and use Bayes' theorem when combining prior prevalence with new evidence. Incorporate distance measures such as Euclidean distance for clustering after scaling the features. Use normalization consistently, with scaling parameters fitted on training data and then applied to validation and test data to prevent data leakage. Select model capacity using validation performance and regularization to balance underfitting and overfitting.

D. Evidence-Based Defence

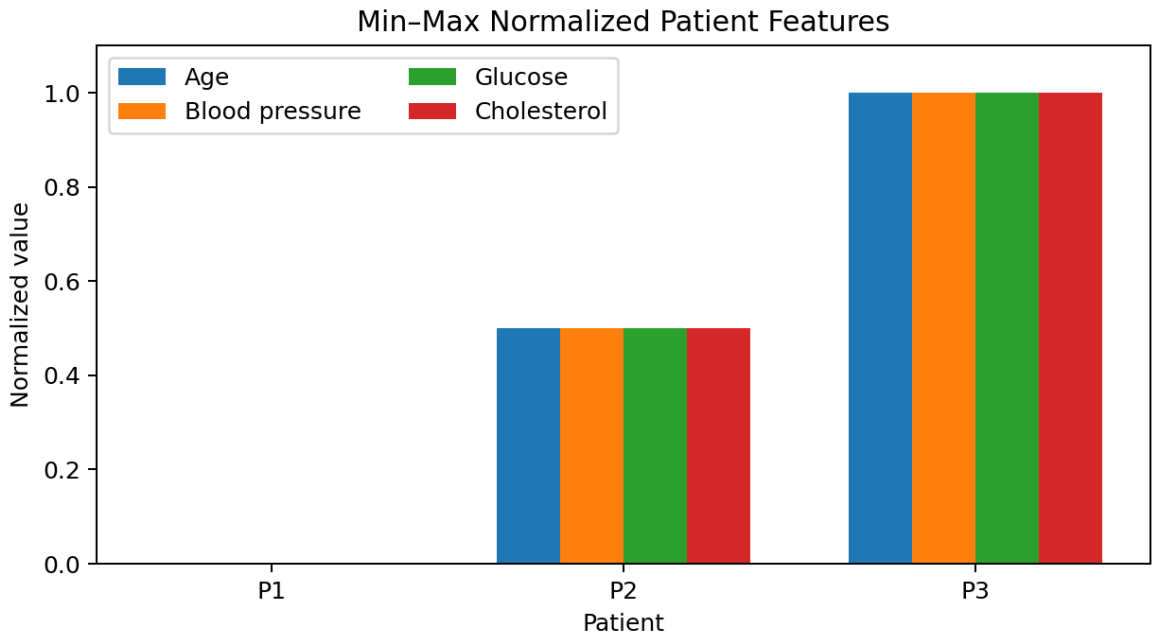
a) Compare the original supervised-only approach with the redesigned classification-plus-clustering strategy using the same data splits and preprocessing rules. For classification, report accuracy, precision, recall/sensitivity, specificity, F1-score, ROC-AUC, confusion matrix and calibration. Because missed disease cases can be serious, examine sensitivity and false-negative rates, not accuracy alone. For clustering, examine silhouette score, within-cluster sum of squares (inertia), cluster sizes and stability across repeated runs; also inspect whether cluster profiles are clinically interpretable. Measure runtime, memory use and scalability on increasing data volumes.

b) The redesigned strategy is justified because it matches each task to an appropriate learning paradigm: supervised classification predicts known disease labels, while unsupervised clustering explores structure in labelled and unlabelled patient features. Normalization makes numerical comparisons meaningful; missing-value handling improves data quality; probability represents uncertainty; and validation with regularization controls model capacity. Experiments on held-out data and scalability tests establish whether the system generalizes and can process one million records efficiently. The system should support, not replace, clinical judgement.

GRAPHICAL REPRESENTATIONS

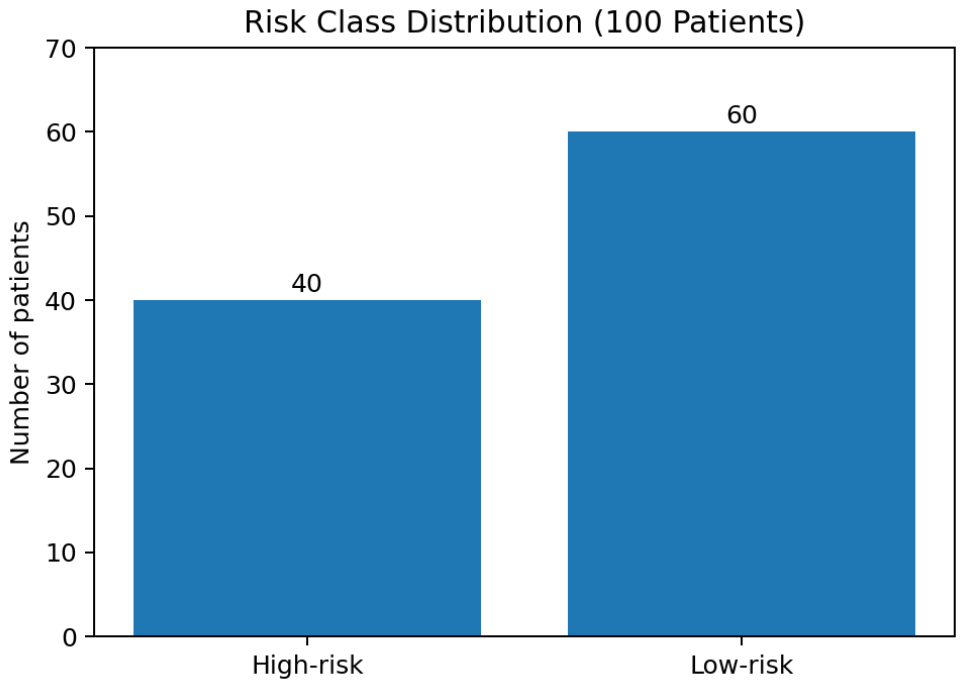
Graph 1 – Normalized Patient Features (Task 1)

The graph displays the min–max normalized values for the three illustrative patients. Each feature is mapped to the same 0–1 scale, making the features easier to compare.



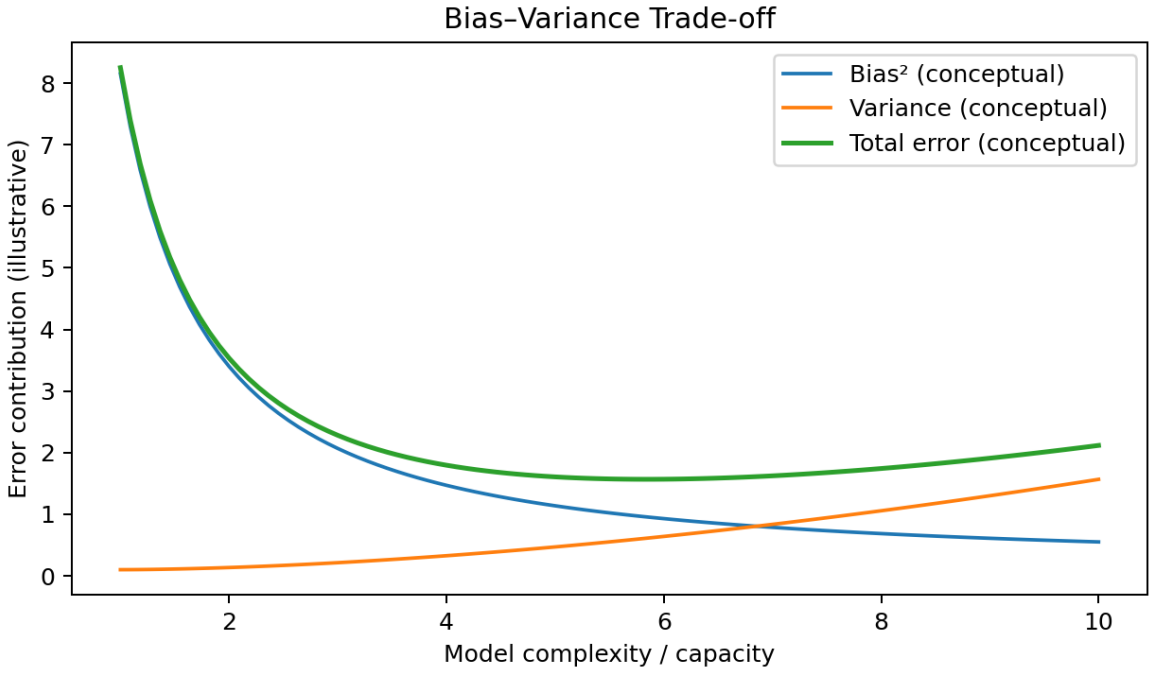
Graph 2 – Patient Risk Class Distribution (Task 5)

The dataset contains 40 high-risk and 60 low-risk records. This 40:60 distribution corresponds to the entropy calculation of approximately 0.971 bits.



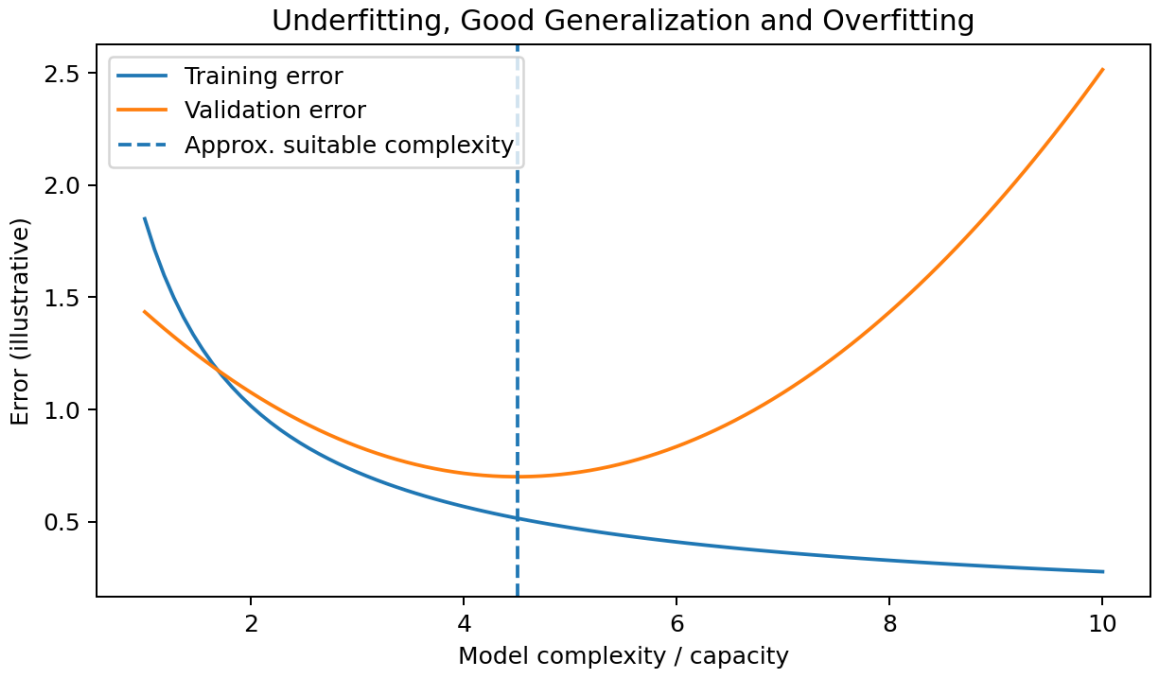
Graph 3 – Bias–Variance Trade-off (Tasks 7 and 8)

This is a conceptual illustration, not measured hospital data. As model complexity increases, bias generally decreases while variance increases. Their combined effect helps explain why a model can become too simple or too complex.



Graph 4 – Model Capacity and Generalization (Tasks 7 and 8)

This conceptual graph shows training error generally decreasing with capacity, while validation error may decrease at first and rise later. The middle region represents a possible balance that generalizes better.



RESULT:

The assignment successfully demonstrates how mathematical foundations and machine learning principles can be applied to healthcare risk prediction. Patient data was represented using a feature matrix and normalized using min–max scaling so that features with different numerical ranges could be compared fairly. Vector operations were used to calculate a weighted preliminary risk score.

The probability analysis showed that 28% of the sample patients had elevated glucose, the probability of disease among patients with elevated glucose was approximately 57.14%, and the probability of elevated glucose among diseased patients was 80%. Using Bayes' theorem, the probability of disease after a positive test was approximately 66.67%. The entropy of the 40:60 risk-class distribution was approximately 0.971 bits.

The Euclidean distance between the two normalized patient records was approximately 0.364. The assignment also explained the bias–variance trade-off, underfitting, overfitting, model capacity, supervised classification and unsupervised clustering. Finally, a redesigned strategy was proposed that combines supervised learning for disease-risk prediction with unsupervised clustering for discovering patient groups, supported by normalization, missing-value handling and suitable evaluation measures.

Therefore, the proposed approach can support the analysis of large healthcare datasets and help identify patients or groups that may require further investigation. The system is intended as a decision-support tool and must not replace professional medical diagnosis.